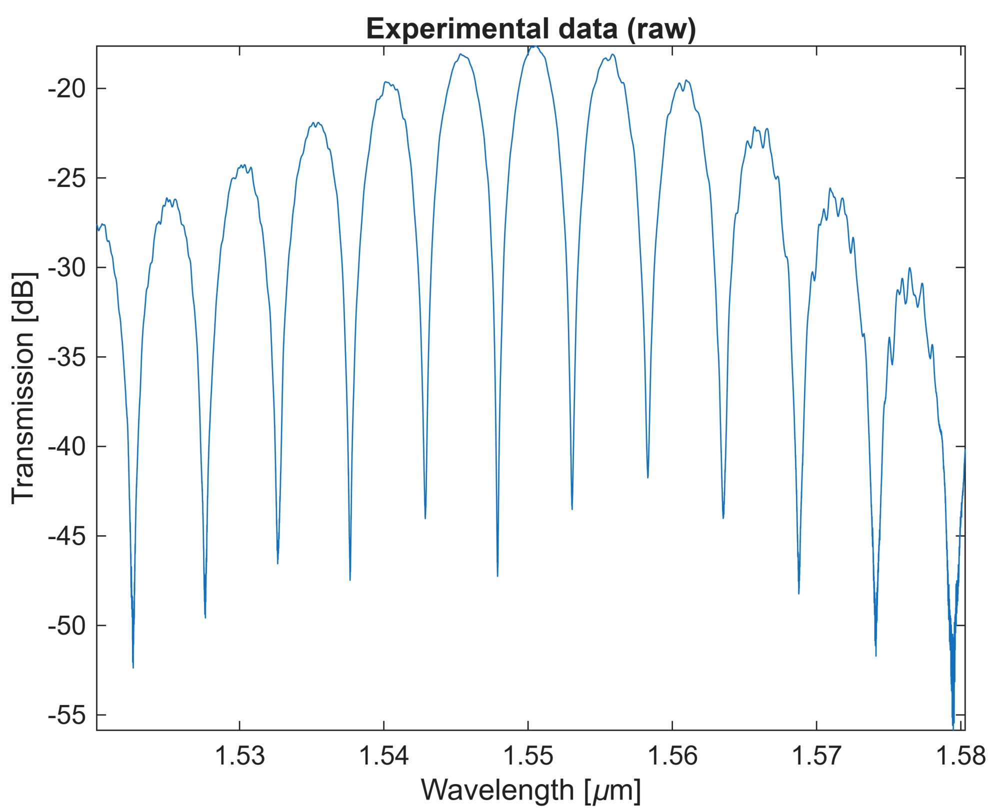
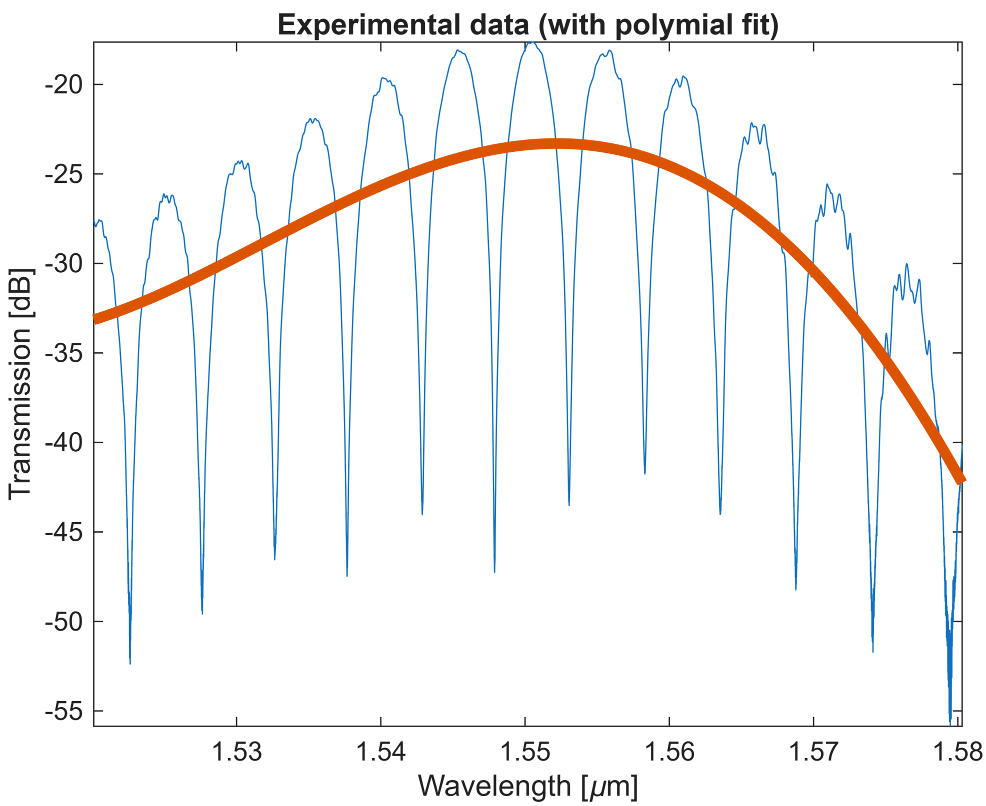
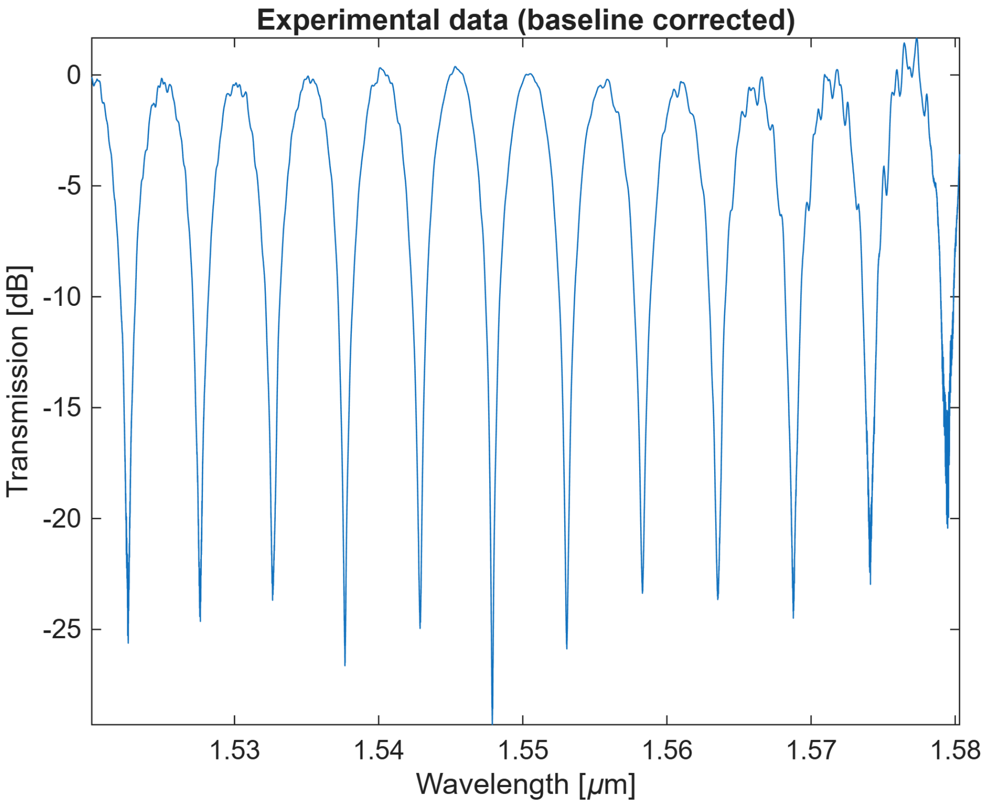
```
% Enter the order of the polynomial for the curve fitting:
POLY_ORDER = 4;

% Enter the Dropbox URL here.  Make sure it has a =1 at the end:
url = 'https://www.dropbox.com/s/1rvjfef4jqybc12/ZiheGao_MZI2_271_Scan1.mat?
dl=1';
%url= 'G:\8_Teaching_SCA\Postgraduate Teaching\Photonic Integrated
Circuits\Unit III\Measurement-Data Analysis\Base Line
Correction\ZiheGao_MZI2_271_Scan1.mat?dl=1' ;
PORT=1; % Which Fibre array port is the output connected to?

a=websave('a.mat',url); % get data from Dropbox
load('a.mat');

% Data is stored in variable "scanResults".
% There are two columns - wavelength (1), and amplitude (2)
lambda=scanResults(1,PORT).Data(:,1)/1e9;
amplitude=scanResults(1,PORT).Data(:,2);

% Plot the raw data:
figure;
plot (lambda*1e6, amplitude);
xlabel ('Wavelength [\mum]');
ylabel ('Transmission [dB]');
axis tight
title ('Experimental data (raw)');
```



```
% Curve fit data to a polynomial for baseline correction
p=polyfit((lambda-mean(lambda))*1e6, amplitude, POLY_ORDER);
amplitude_baseline=polyval(p,(lambda-mean(lambda))*1e6);
figure;
plot (lambda*1e6, amplitude); hold all;
plot (lambda*1e6, amplitude_baseline, 'LineWidth',4);
xlabel ('Wavelength [\mum]');
ylabel ('Transmission [dB]');
axis tight
title ('Experimental data (with polymial fit)');
```




```
% Perform baseline correction to flatten the spectrum
% Use the curve polynomial, and subtract from original data
amplitude_corrected = amplitude - amplitude_baseline;
amplitude_corrected = amplitude_corrected + max(amplitude_baseline) -
max(amplitude);
figure;
plot (lambda*1e6, amplitude_corrected);
xlabel ('Wavelength [\mum]');
ylabel ('Transmission [dB]');
axis tight
title ('Experimental data (baseline corrected)');
```


Experimental data (baseline corrected)
Transmission [dB]
0
-5
-10
-15
-20
-25
1.53
1.54
1.55
1.56
1.57
1.58
Wavelength [μm]